

Variable Selection and Class-Only MNARz Clustering with SelvarMixMNAR

SelvarMixMNAR is the direct R implementation accompanying the NeurIPS 2025 paper *A Unified Framework for Variable Selection in Model-Based Clustering with Missing Not at Random* [1]. The package combines a penalized Gaussian-mixture ranking, SRUW variable roles, and a class-only MNARz missingness model.

For incomplete data, `decoupled_mnarz` estimates SRUW roles from a completed working matrix before fitting an MNARz mixture, whereas `joint_mnarz` estimates the SRUW and missingness terms under one observed-data criterion.

1 SRUW roles and the fitted MNARz mechanism

For observation $Y_i \in \mathbb{R}^p$, the SRUW factorization uses three disjoint role sets and one predictor subset:

- S : variables that define the Gaussian mixture;
- $R \subseteq S$: clustering variables used as predictors of redundant variables;
- U : redundant variables modeled by a common Gaussian regression on Y_i^R ;
- W : variables modeled independently of the clustering block.

Writing K for the number of mixture components, the factorization has the form

$$f(Y_i) = f_{\text{clust}}(Y_i^S \mid K, m) f_{\text{reg}}(Y_i^U \mid Y_i^R, r) f_{\text{indep}}(Y_i^W \mid l).$$

Here m , r , and l index the covariance models for the clustering, regression, and independent blocks, respectively.

The fitted missingness class is MNARz. If $C_{ij} = 1$ denotes a missing entry and $Z_i = k$ denotes latent component membership, then

$$\Pr(C_{ij} = 1 \mid Z_i = k) = \rho_k.$$

Thus each component has one missingness probability shared across variables. MNARy, in which missingness depends directly on an unobserved value, and MNARzj, in which the class effect varies by coordinate, require different likelihoods. Identifiability relies on the MNARz assumptions stated in the accompanying paper in addition to numerical convergence.

The separate decoupled fit also permits a prespecified mixed mask through `mnarz_control = list(mecha = "mixed", is_mnar = m)`. All coordinates with $m_j = \text{TRUE}$ share the class-specific parameter ρ_k . Mask factors with $m_j = \text{FALSE}$ are treated as ignorable and are omitted from the modeled likelihood. The vector m is supplied by the analyst, and all selected coordinates share ρ_k . It is therefore a class-only sensitivity specification rather than MNARzj.

2 Two stages of variable selection

The ranking stage fits penalized Gaussian mixtures over a grid of mean and precision penalties. A variable receives a high score when its component means remain nonzero across many valid grid fits. Each candidate value of K owns its initializer, penalty grid, fitted states, and failure records.

The role-selection stage scans the ranked variables. A variable enters S when its clustering contribution improves the model criterion. Variables not retained in S are assigned to U or W by comparing common regression and independent Gaussian models. The stopping parameter `hsize` is the number of consecutive nonpositive criterion increments tolerated before the scan stops; an improvement resets the count.

3 A seeded complete-data example

The following Gaussian example has two clustering variables, two redundant variables, and two independent variables. The generating labels are retained only for external evaluation and are not supplied to the estimator.

```
library(SelvarMixMNAR)
set.seed(2025)
n <- 90
latent_class <- rep(1:3, each = n / 3)
signal_1 <- rnorm(n, c(-2, 0, 2)[latent_class], 0.7)
signal_2 <- rnorm(n, c(1.5, -1.5, 0)[latent_class], 0.7)
x <- cbind(
  signal_1 = signal_1,
  signal_2 = signal_2,
  redundant_1 = 0.8 * signal_1 + rnorm(n, sd = 0.35),
  redundant_2 = -0.7 * signal_2 + rnorm(n, sd = 0.35),
  noise_1 = rnorm(n),
  noise_2 = rnorm(n)
)
dim(x)

#> [1] 90 6
```

The known variable order is supplied through `rank` to isolate the SRUW role-selection stage in this synthetic example. Omitting `rank` estimates the penalized ordering from the data.

```
set.seed(2026)
fit <- SelvarClustLasso(
  x = x,
  nbcluster = 3,
  models = "VVI",
  rank = seq_len(ncol(x)),
  hsize = 2,
  nbcores = 1,
  workflow = "imputed_sruw",
  init_control = list(method = "mclust_hc", seed = 2026),
  selection_control = list(stopping = "consecutive", seed = 2026)
)
```

The `imputed_sruw` workflow fits the ranking and SRUW role model on the complete working data without a missingness likelihood. The default Mclust covariance model is VVI, a diagonal model with component-specific volume and shape.

4 Decoupled and joint MNARz workflows

When `x` is incomplete, `workflow = "auto"` resolves to `"decoupled_mnarz"`. Table 1 separates the three statistical targets.

Table 1: Workflow-specific statistical targets.

Workflow	Target
<code>imputed_sruw</code>	SRUW ranking and role selection on a completed working data set; no missingness likelihood.
<code>decoupled_mnarz</code>	Imputed-data SRUW selection followed by a separate, role-agnostic MNARz Gaussian mixture.
<code>joint_mnarz</code>	Observed-data SRUW and MNARz estimation on the original incomplete matrix, with forward and reverse role updates scored by total observed-data BIC.

The decoupled estimator reduces computation by separating imputed-data role selection from the missingness fit. The joint estimator couples both terms in the observed-data objective and consequently requires more repeated mixture fits along the role path.

```
set.seed(2027)
x_miss <- x
rho <- c(0.02, 0.08, 0.15)
mask_probability <- matrix(
  rho[latent_class], nrow = nrow(x_miss), ncol = ncol(x_miss)
)
mask <- matrix(runif(length(x_miss)), nrow = nrow(x_miss)) < mask_probability
x_miss[mask] <- NA_real_

common <- list(
  nbcluster = 2:4,
  models = "VVI",
  num_vals_penalty = 3,
  hsize = 2,
  nbcores = 1,
  init_control = list(method = "mclust_hc", seed = 2027),
  selection_control = list(stopping = "consecutive", seed = 2027)
)

fit_decoupled <- do.call(
  SelvarClustLasso,
  c(list(x = x_miss, workflow = "decoupled_mnarz"), common)
)

fit_joint <- do.call(
  SelvarClustLasso,
  c(list(x = x_miss, workflow = "joint_mnarz"), common)
)
```

Ranking requires one complete working data set. With `use_copula = TRUE`, the package uses `gcimputeR`; setting `use_copula = FALSE` requests `missRanger`. The imputation supplies the ranking stage in both MNARz workflows. The joint fit itself returns to the original incomplete matrix.

5 Initialization and warm paths

Initialization is data-driven and performed separately for each candidate component count. The control `init_control$method` accepts the following values:

- `mclust_hc` and `mclust_em`;
- `kmeans` and `random`;
- `user` and `previous_fit`;
- `deterministic_multistart`.

Every route produces the same numeric state: component proportions, means, covariances, precisions, responsibilities, partition, objective, method, seed, and failure status.

Independent cold starts are the ranking default. Warm starts are requested through `rank_control$warm_start`:

```
fit_warm <- SelvarClustLasso(
  x,
  nbcluster = 2:4,
  workflow = "imputed_sruw",
  nbcores = 1,
  rank_control = list(warm_start = "both")
)
```

An inner warm start continues the graphical-lasso solver between EM iterations. An outer warm start passes the complete fitted mixture state between adjacent points of a deterministic penalty path. Points within one outer path are serial because each depends on its predecessor; parallelism is reserved for independent component counts or initializations.

6 Results and diagnostics

A fitted `selvarmix` object provides standard methods:

```
fit
summary(fit)
plot(fit)
```

The default plot displays the counts in the $S/U/W$ role partition; R is reported separately because it is a subset of S . The result also contains the selected component count, criterion metadata, partition, posterior probabilities, Gaussian and regression parameters, preprocessing transformations, and the finite completed working-data matrix. Workflow-specific fields preserve the MNARz or joint fit, initialization provenance, penalty grids, convergence status, and role-selection stopping diagnostics.

Failed ranking fits remain visible in the recorded grid state. Fits with non-finite objectives, invalid component masses, non-positive-definite covariances, inconsistent covariance and precision matrices, material objective decreases, or unsuccessful graphical-lasso termination are not scored. This rule prevents numerical failure from appearing as evidence that a variable has been shrunk to zero.

7 Bioinformatics interpretation

For a continuous molecular-assay matrix, S identifies measurements that define latent subgroups, U contains correlated measurements explained by the predictor subset R , and W contains measurements modeled independently of the clustering block. Class-only MNARz represents subgroup-dependent observation rates when latent biological states differ in overall detectability.

The Gaussian likelihood presumes that library-size normalization, variance stabilization, batch correction, and domain-specific filtering have already been performed. Abundance-dependent dropout and feature-specific detection probabilities require a different missingness model.

8 Interpretation and limitations

1. Gaussian finite-mixture likelihoods are non-convex. The returned fit is the local solution associated with the requested initialization and model path.
2. Numeric component labels are arbitrary. Comparisons across fits should use label-invariant summaries or an explicit label alignment.
3. The MNARz mechanism assumes one component-specific missingness probability across coordinates. A scientific application should justify this restriction from its data-collection process.
4. The optional mixed mask is fixed by the analyst for the separate decoupled fit, and estimation conditions on that partition of the coordinates.
5. `joint_mnarz` requires more computation than `decoupled_mnarz`. Comparisons of clustering, selection, and imputation accuracy should use replicated designs matched to the scientific setting.
6. Mclust and Rmixmod are the supported SRUW backends. MixAll tokens stop before native execution because an STK failure may terminate the R session; Mclust VVI is the default.
7. Convergence, KKT, and conditioning diagnostics characterize the returned numerical solution. Global optimality, BIC consistency, and selection consistency require separate theoretical conditions.

References

- [1] Binh H. Ho, Long Nguyen Chi, TrungTin Nguyen, Binh T. Nguyen, Van Ha Hoang, and Christopher Drovandi. A unified framework for variable selection in model-based clustering with missing not at random. In *Advances in Neural Information Processing Systems*, 2025. 39th Conference on Neural Information Processing Systems (NeurIPS 2025).